Cross-Task Meta-Learning Network With Graph-Enhanced Attention Module for Hyperspectral Change Detection

Rui Miao, Yuxiang Zhang, *Senior Member, IEEE*, Yanni Dong, *Senior Member, IEEE*, and Bo Du, *Senior Member, IEEE*

***Abstract*— Hyperspectral change detection (HCD) with limited training samples has attracted increasing attention in recent years. The current research works require to train a dedicated model for each dataset, resulting in limited generalization. Moreover, for the same dataset, a model that has been trained for binary change detection (CD) task usually needs to be retrained for multiclass CD task, thereby leading to low efficiency. We propose a cross-task meta-learning network with graph-enhanced attention module (CMGA) for both binary CD tasks and multiclass CD tasks, aiming to fully extract the spatial–spectral features of hyperspectral images (HSIs) and efficiently transfer the prior knowledge to accomplish various HCD tasks with limited training samples. In order to capture general prior knowledge, the multitask meta-optimization is designed to train the pretraining model with the ability of addressing cross-task new classes. Moreover, the diverse tasks are unified into one framework, which aims at identifying the target class from other classes to improve the model efficiency and generalization. The graph-enhanced attention (GA) module is developed to leverage the graph knowledge extract from the graph transformer (GT) module to enhance the pixel-level feature representation. Experiments on four hyperspectral datasets illustrated the effectiveness of the proposed algorithm.**

***Index Terms*— Change detection (CD), graph transformer (GT), hyperspectral images (HSIs), meta-learning, transfer learning.**

I. Introduction

With high spectral resolution, hyperspectral image (HSI) is able to reflect the fine spectral composition of objects and has made advanced development in Earth observation research, such as hyperspectral classification [1] and change detection (CD) [2]. Among these researches, hyperspectral change detection (HCD) has attracted increasing attention and has been widely used in reality scenarios, such as land cover mapping, agriculture and forestry detection, and resource exploration [3], [4], [5]. HCD aims to detect the change information in multitemporal HSIs [6], which can be roughly divided into the fundamental hyperspectral binary CD (HBCD) [7] and the challenging hyperspectral multiclass CD (HMCD) [8]. Compared to HBCD focusing on detecting the changed areas and the unchanged areas, HMCD also requires to detect various change classes rather than a single change. In order to detect subtle changes and distinguish various change classes, it is critical to make full use of the abundant spectral–spatial–temporal information in multitemporal HSIs.

With the strong ability of learning rich features, deep learning (DL) methods have gained significant momentum in various fields [9], [10]. The convolutional neural network (CNN) and the transformer among various DL architectures have gained widespread popularity in image processing [11], [12], [13], [14]. For example, Wang et al. [15] presented an end-to-end 2-D CNN framework to fuse multisource information at both pixel and subpixel levels for detecting the changed regions. Qu et al. [16] proposed a multilevel encoder–decoder attention network (ML-EDAN) to efficiently use the hierarchical features for detecting the changed areas. Luo et al. [17] proposed the multiscale diff-changed feature fusion network (MSDFFN), aiming to detect the subtle changed areas between bitemporal HSIs on different scales. Song et al. [18] utilized the recurrent 3-D fully convolutional network (Re3FCN) to distinguish various change classes. Ge et al. [19] developed a multidirection and multiscale spectral–spatial residual network to efficiently capture the spectral–spatial features for detecting changed areas. Wang et al. [20] designed a global fully convolutional network with spatial attention mechanism to extract features for distinguishing various change classes. Hu et al. [6] proposed a binary change-guided network (BCG-Net), which can use binary CD result computed by spectral unmixing result to guide multiclass CD. The transformer-based model is capable of effectively modeling global dependencies within sequential data, thus attaining competitive performance in the detection of changed areas. For example, Wang et al. [21] developed a spectral–spatial–temporal transformer (SST-Former) for detecting changed areas, focusing on the spatial–spectral information and the significant temporal sequences of bitemporal HSIs. Tang et al. [22] proposed a W-shaped dual-Siamese branch hierarchical

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network (WNet) to fully extract local–global features by combining CNN and transformer. Although the abovementioned methods show good performance in detecting changed regions and distinguishing various change classes, these methods usually require a considerable amount of training samples to train the network. However, annotating sufficient training samples will result in significant time and labor costs.

To address the issue of insufficient labeled training samples, some approaches are proposed and demonstrated the excellent performance with limited labeled training samples. One typical approach is to leverage the message passing mechanism of graph neural networks (GNNs) to make full use of the labeled and unlabeled samples [23], [24], [25], [26], [27]. With the advantage of processing irregular data, GNNs have been widely used in HIS processing to capture the complex structural properties of objects [28], [29], [30], [31], [32]. For example, Qu et al. [33] developed the dual-branch difference amplification graph convolutional network (D^2AGCN) to extract the superpixel-level features for detecting changed areas. Wang et al. [34] proposed the dual-branch framework based on temporal–spatial joint graph attention with complement strategy (CSDBF) to detect changed areas. Zhang et al. [35] proposed a multiorder graph convolutional network (MGCN) for efficiently detecting changed areas pixel by pixel. Dong et al. [36] developed a dual-branch local information-enhanced graph-transformer (D-LIEG) network to detect changed areas and distinguish various change classes, which extracts the local information by a convolutional layer to enhance the local information of each graph transformer (GT) block. Another typical approach is to extract domain-aligned priori knowledge to help models to effectively learn significant features. For example, Wang et al. [37] treated the bitemporal HSIs as two different image domains with overlapping features, utilized a powerful feature extractor to align the domains by many unlabeled data, and then separated different features by few labeled data. Qu et al. [38] proposed a cycle-refined multidecision joint alignment network (CMJAN) to learn changes from source domain and adapt them into target domain. The data distributions between source and target domains are progressively aligned in CMJAN. A feature mutual representation-based graph domain adaptive network (FGDANet) [39] was proposed by using a graph domain adaptation strategy and progressive graph-to-pixel joint constraint strategy to improve the performance of detecting changed areas. Besides, some approaches [40], [41], [42] utilized the sample generation strategy to generate massive pseudo-labels for the training process. A large number of high-quality pseudo-labels are helpful to effectively train models, resulting in improved CD performance. To obtain multiclass change results, the aforementioned approaches should simultaneously distinguish between changed and unchanged regions, as well as discriminate various change classes by a multiclass classifier, which may obtain a much worse performance for the HMCD task than that for the HBCD task. The existing methods require to train a dedicated model for each HBCD or HMCD task, resulting in limited efficiency and generalization.

Recently, meta-learning providing models of high generalization has been successfully applied in HSI processing domains, including classification and target detection [43], [44], [45], [46], [47]. By teaching models how to learn, meta-learning can help models to efficiently adapt to new tasks or domains, thereby improving the model generalization [48]. Inspired by these ideas, we propose a cross-task meta-learning network with graph-enhanced attention module (CMGA) for HCD, aiming to improve the model efficiency and generalization. The main idea of CMGA method is to transfer the meta-knowledge captured from hyperspectral classification tasks to guide various HCD tasks. The overall architecture comprises three aspects: graph-enhanced attention (GA) module, multitask meta-optimization, and meta-knowledge-guided CD. First, the GA module is proposed for both multitask meta-optimization and meta-knowledge-guided CD. For multitask meta-optimization, dual-branch GA module integrates the GT module and pixel-level multiscale attention (PMA) module to ensure comprehensive consideration of superpixel- and pixel-level features. For meta-knowledge-guided CD, the three-branch GA module utilizes two GT modules to extract the superpixel-level features of bitemporal HSIs and introduces a PMA module to extract the pixel-level features of the subtraction of bitemporal HSIs. Second, the multitask meta-optimization framework is designed to pretrain a two-branch GA module to capture general meta-knowledge by training it to handle multiple hyperspectral classification tasks. Considering the rich variety of classes in hyperspectral classification datasets, we utilize them to set up multiple binary classification tasks in multitask meta-optimization. The goal of each task is to identify the target class from the other class, aiming to capture meta-knowledge to differentiate between any class effectively. Finally, the meta-knowledge is transferred to effectively guide various CD tasks by freezing parameters and only training few parameters with limited labeled training samples. It is worth noting that the CMGA method can implement both HBCD and HMCD tasks. Although the number of classes varies, the essence of HBCD and HMCD is to differentiate between different classes. Therefore, we unify HBCD and HMCD into one framework by identifying the target class once from other classes progressively and determining the final class of each pixel based on a scoring system. Our main contributions are summarized as follows.

1) A GA module based on GT and PMA is proposed to fully extract the spatial–spectral features at both superpixel and pixel levels.
2) A multitask meta-optimization framework is designed to pretrain GA module, aiming to learn the ability of differentiating between different classes and obtain the general meta-knowledge from classification tasks.
3) A unified transfer framework is proposed to implement HBCD and HMCD tasks, which can leverage the obtained meta-knowledge to guide various new tasks with limited labeled training samples.

The reminder of this article is organized as follows. The related work is analyzed in Section II. Details of the proposed method are described in Section III. The experimental results and discussions are presented in Section IV. Finally, conclusions are drawn in Section V.

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II. Related Work

Meta-learning is a machine learning approach with the thought of "learn-to-learn." Instead of the traditional train–test paradigm, meta-learning focuses on training models that can generalize well to new tasks by leveraging knowledge gained from training on diverse meta-tasks. The meta-training phase involves two stages: within-task training with support set and within-task testing with query set. The model parameters are updated by calculating and minimizing the average loss between the predicting labels and real labels of the query samples [49]. Meta-learning methods can be roughly divided into three categories: metric-based methods [50], model-based methods [51], and optimization-based methods [52].

In the field of hyperspectral processing, researchers have mainly favored the meta-learning methods based on metric learning for few-shot HSI applications [43]. A critical step in these methods is the calculation of the distance between features in the source domain and target domain. Then, domain adaptation is applied to enable the model to achieve the generalization from the source domain to target domain. For example, Liu et al. [53] proposed a refined prototypical contrastive learning method for few-shot HSI classification, named RPCL-FSL. In RPCL-FSL, triple constraints, including contrastive learning-based constraint, self-calibration-based constraint, and cross-calibration-based constraint, were imposed to stabilize and refine the prototypes. Also, a fusion training strategy was designed to align the source domain and target domain, thereby alleviating the domain shift problem. In HCD, Wang et al. [54] proposed the cross-heterogeneous domain few-shot learning framework based on bidirectional generation (BiG-FSLF). BiG-FSLF introduced metric-based meta-learning to solve the limited sample issue in HCD. A very high-resolution CD dataset is used as the source domain dataset, and a different hyperspectral CD dataset is used as the target domain dataset. To align the feature distributions of two heterogeneous domains, BiG-FSLF requires to train a specific model for each hyperspectral CD dataset. Therefore, the existing methods are still insufficient in model efficiency and generalization when facing hyperspectral CD datasets with diverse distributions.

Differently, we propose a cross-task meta-learning network with graph-enhanced attention module (CMGA) for HCD, which introduces optimization-based meta-learning to update the model parameters by setting various binary classification tasks with the same goal, thereby obtaining the meta-knowledge suitable for CD tasks. By unifying the goals of meta-optimization and HCD tasks to detect the target class, the optimized model can be efficiently transferred to accomplish HCD tasks. As far as we know, this is a new attempt to accomplish HBCD and HMCD tasks with limited labeled samples in a high generalization and efficiency way.

III. Methodology

CMGA mainly consists of two processes including multitask meta-optimization and meta-knowledge-guided CD. A classification network with dual-branch GA module is designed for the former process and a CD network with three-branch GA module is designed for the latter process. The overall architecture of the proposed CMGA is shown in Fig. 1.

1) Notably, the GA module, which comprises of GT module and PMA module, is designed to extract the valuable superpixel-level and pixel-level features during the two processes. For the latter process, two identical GT modules are used for bitemporal images and one PMA module is used for the differential image, thereby forming a three-branch GA module.
2) In the multitask meta-optimization process, the dual-branch GA module is trained by multiple binary classification tasks, aiming to learn the ability of identifying different classes, thereby obtaining the general meta-knowledge $\widehat{\phi}$. Specifically, the meta-training datasets with massive labeled samples are applied to construct T binary classification tasks for within-task training and with-task testing. The support data χ^{sup} of within-task training are utilized to train the classification network f_ϕ, and the query data χ^{que} of within-task testing are then utilized to validate the ability of binary classification.
3) In the meta-knowledge-guided CD process, by transferring the trained dual-branch GA module with frozen parameters $\widehat{\theta}$, a CD framework based on a three-branch GA module is designed to effectively detect changed areas and distinguish various change classes with limited training samples. The HBCD and HMCD tasks are unified into one framework by progressive detection strategy, leading to high efficiency and generalization. Specifically, each change class is detected by a fine-tuned CD network, and then, the scoring system is applied to obtain the final multiclass change results. In order to improve the performance of CD, a cosine similarity loss function is designed to amplify the feature differences in bitemporal HSIs.

A. Notation

We denote the training dataset in the multitask meta-optimization as $\boldsymbol{HSI}_{\text{tra}} \in \mathbb{R}^{h\times w\times b}$, h, w, and b denote height, weight, and bands of the meta-training HSI data, respectively. We denote the meta-testing dataset in the meta-knowledge-guided CD as $\boldsymbol{HSI}_{\text{tes}} = (\boldsymbol{T}_1 \in \mathbb{R}^{H\times W\times B}, \boldsymbol{T_2} \in \mathbb{R}^{H\times W\times B})$, where H, W, and B denotes height, weight, and bands of the meta-testing HSI data, respectively. For the dual-branch GA module, the input of GT is represented as $\boldsymbol{G}_0 = (\boldsymbol{A}_0, \boldsymbol{H}_0)$, and the input of PMA is represented as $\boldsymbol{X}_{\text{tra}}$. For the three-branch GA module, the input of GT is represented as $\boldsymbol{G}_1 = (\boldsymbol{A}_1, \boldsymbol{H}_1)$ and $\boldsymbol{G}_2 = (\boldsymbol{A}_2, \boldsymbol{H}_2)$ for bitemporal HSIs, and the input of PMA is represented as $\boldsymbol{X}_{\text{tes}} = \boldsymbol{T}_1 - \boldsymbol{T}_2$. The final output of GT and PMA modules is denoted as $\boldsymbol{Spf}$ with d channels and $\boldsymbol{Pf}$ with C channels, respectively. The three-branch GA module will output two superpixel-level features at pixel space for bitemporal HSIs, which can be denoted as $\boldsymbol{Z}_1$ and $\boldsymbol{Z}_2$, respectively. In the multitask meta-optimization, the model used is denoted as f_ϕ. In the meta-knowledge-guided CD task, the initial parameters $\widehat{\phi}$ of model are obtained by multitask

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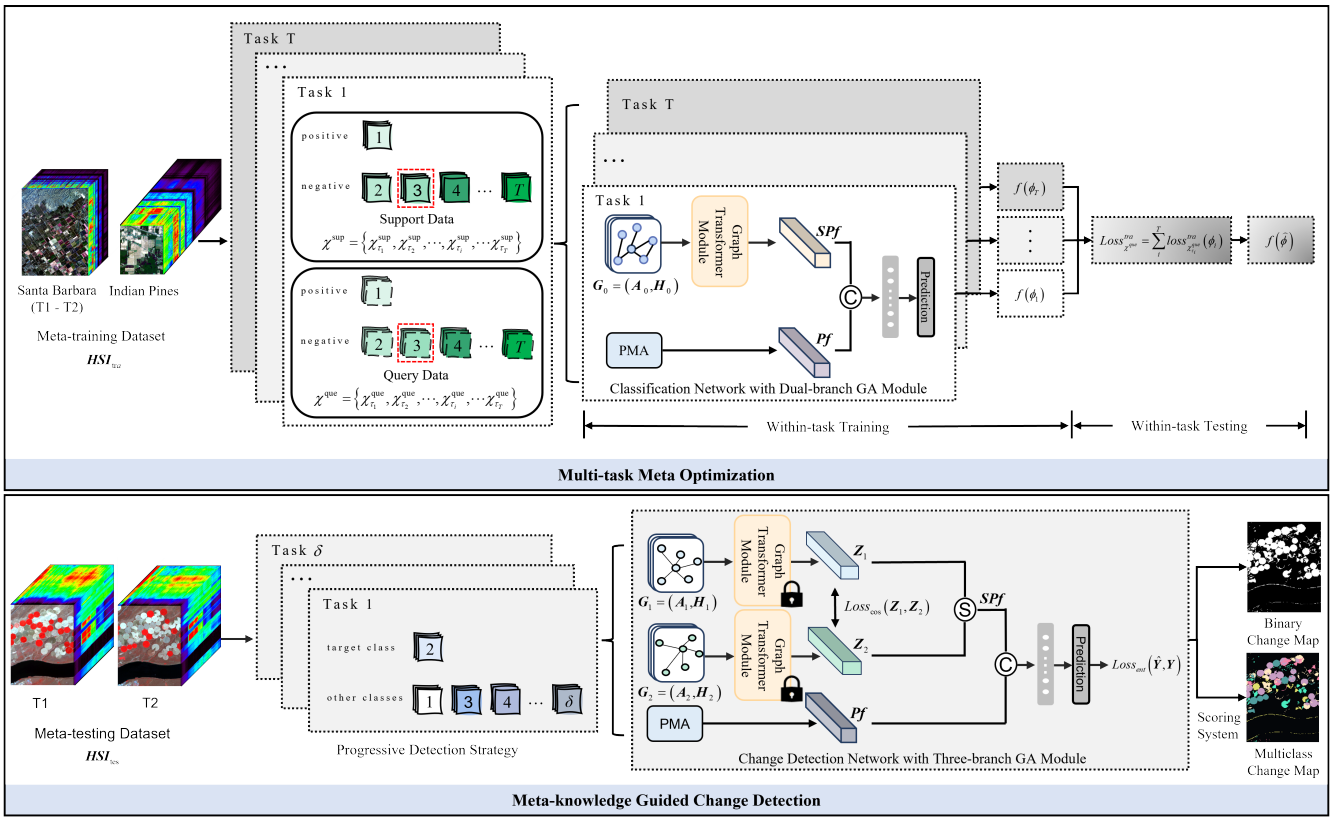


Fig. 1. Overall architecture of the proposed CMGA. The goal of multitask meta-optimization is to obtain general meta-knowledge by handling multiple binary classification tasks. Each task contains positive samples with a label value of i and negative samples with a label value j (j is not equal to i). Then, by progressively detecting change class from other classes, the meta-knowledge is transferred to effectively guide various change detection tasks with limited training samples.

meta-optimization, and the parameters $\widehat{\theta}$ of GT module are frozen in this process.

B. Graph-Enhanced Attention Module

1) Pixel-Level Multiscale Attention Module: The PMA module combines multiscale CNN and channel attention module [26] to fully extract the spatial–spectral features at the pixel level. CNN adopts patch convolution to extract the spatial–spectral features of regular areas with fixed size [55]. To better extract the local features, the PMA module utilizes multiscale CNN to dynamically integrate path convolutions from regular areas with different sizes. In order to effectively capture abundant spectral information from HSI, a channel attention module is used to update the features extracted by multiscale CNN. The details of PMA module are illustrated in Fig. 2.

Given an HSI cube $\boldsymbol{X}$ for PMA, which can be $\boldsymbol{X}_{\text{tra}} \in \mathbb{R}^{h\times w\times b}$ or $\boldsymbol{X}_{\text{tes}} \in \mathbb{R}^{H\times W\times B}$, a 1×1 convolutional layer is utilized to diminish the channel dimension, thereby reducing the computational complexity and standardizing the number of bands from different data, making the network more uniform and flexible. Then, the dynamic fusion of output features from 1×1, 3×3, and 5×5 convolutional layers is achieved by adding the features with learnable weights. The fused pixel-level features $\boldsymbol{X}^*$ are updated through the channel attention module. The attention map $\boldsymbol{M} \in \mathbb{R}^{C\times C}$ for channels is calculated by the fused features. By multiplying attention map $\boldsymbol{M}$ with $\boldsymbol{X}^*$, the important information of the fused pixel-level features $\boldsymbol{X}^*$ is highlighted. Therefore, the fused features $\boldsymbol{X}^*$ are updated by adding the highlighted features with the original fused features to obtain final pixel-level spatial–spectral features $\boldsymbol{Pf}$. The mathematic expression of PMA module can be defined as follows:

$$\boldsymbol{Pf} = \text{CAM}\left(\sum_{k\in\{1,3,5\}} W_k \cdot f_{\text{Conv}}^{k\times k}\left(f_{\text{Conv}}^{1\times 1}(\boldsymbol{X})\right)\right) \tag{1}$$

where CAM$(\cdot)$ denotes the operation of channel attention module and W_k is the learnable weight coefficient for the features obtained by $k \times k$ convolution layer.

2) Graph Transformer Module: HSI is often composed of many irregular objects with dynamic boundaries and complex spatial relationships, thereby forming a topological structure. In comparison to CNNs and transformer networks, GT networks are flexible in handling data of topological structure [56], [57]. To better preserve the topological structure of HSI, the superpixel graph structure is first constructed by superpixel segmentation and then input into the GT module. The GT module is composed of several GT blocks with self-attention mechanisms and residual connection. It can be used as a single branch for multitask meta-optimization and a dual branch for HCD tasks. Each graph node can dynamically interact with connected nodes through the GT module,

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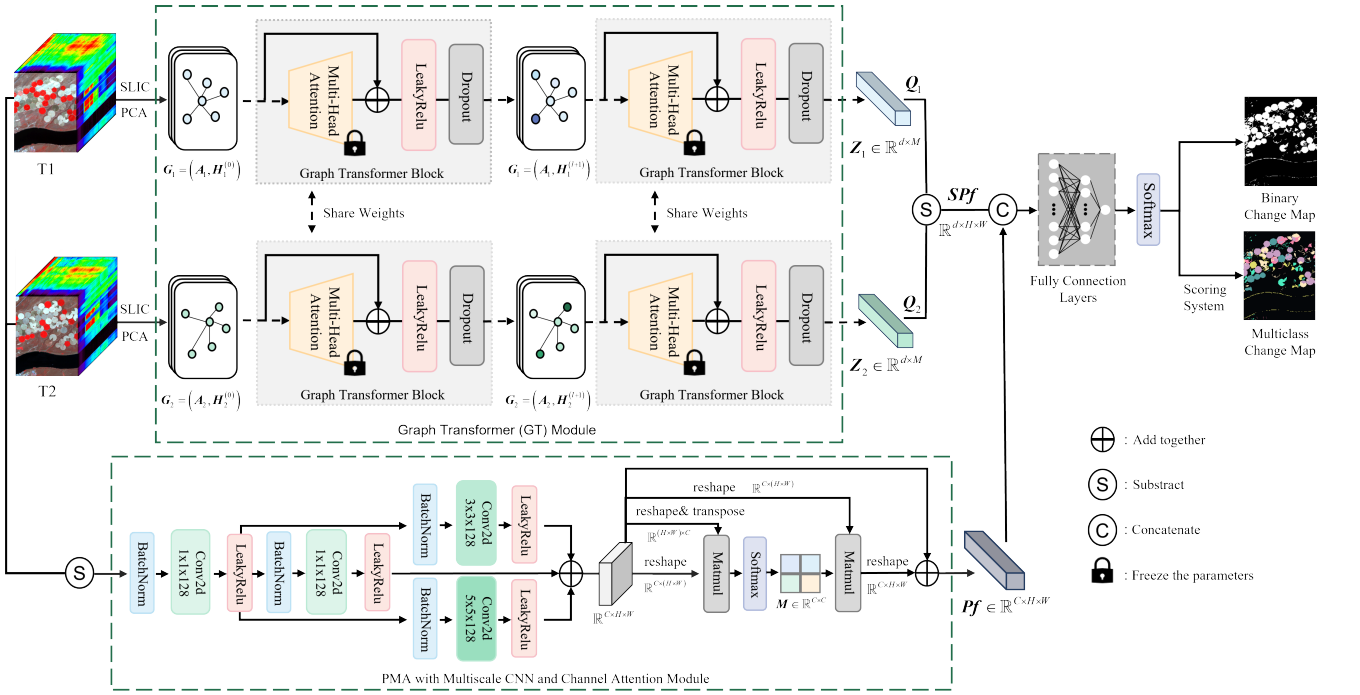


Fig. 2. Detailed meta-knowledge guided change detection network with three-branch GA module. Based on a three-branch GA module, the change detection network is comprised of a GT module with frozen parameters, a PMA module, and fully connection layers.

effectively capturing the implicitly topological structure features of HSI. The GT block is shown in Fig. 2.

The HSI is treated as a superpixel graph $\boldsymbol{G} = (\boldsymbol{A}, \boldsymbol{H})$. By applying simple linear iterative clustering (SLIC) [58], the adjacency matrix $\boldsymbol{A} \in \mathbb{R}^{M\times M}$ and the feature matrix $\bar{\boldsymbol{H}} \in \mathbb{R}^{M\times B/b}$ are obtained from the superpixel segmentation results. $\boldsymbol{A} \in \mathbb{R}^{M\times M}$ is used to represent the connectivity between superpixels. To unify the number of bands between different datasets, PCA is utilized to select the first principal component and obtain the features $\boldsymbol{H} \in \mathbb{R}^{M\times 1}$, which contains most of the information of $\bar{\boldsymbol{H}}$. Therefore, the GT module focuses on the significant spatial features of diverse HSIs. Besides, the association matrix $\boldsymbol{Q} \in \mathbb{R}^{N\times M}$ [59] is used to transform superpixel-level features to pixel-level features. $\boldsymbol{A}$ and $\boldsymbol{H}$ can be defined as follows:

$$\begin{aligned} \boldsymbol{A}_{i,j} &= e^{\frac{\left\|\boldsymbol{h}_i - \boldsymbol{h}_j\right\|^2}{\sigma^2}} \\ \boldsymbol{h}_i &= \mathrm{PCA}\left(\frac{1}{K_i}\sum_{j=1}^{K_i}\boldsymbol{x}_{i,j}\right) \end{aligned} \tag{2}$$

where $\boldsymbol{h}_i$ is the ith element of $\boldsymbol{H}$, representing the feature vector of the ith superpixel, and K_i and $\boldsymbol{x}_{i,j}$ represent the number of pixels and the jth pixel within the ith superpixel, respectively. $\boldsymbol{A}_{i,j}$ represents the connectivity between nodes i and j, and $\sigma = 0.1$. $\boldsymbol{Q}_{i,j} = 0$ denotes that the ith pixel does not belong to the jth superpixel, and $\boldsymbol{Q}_{i,j} = 1$ denotes that the ith pixel belongs to the jth superpixel. N and M denote the total number of pixels and superpixels, respectively; PCA$(\cdot)$ is the process of selecting the first principal component to obtain the feature vector.

For edge from node i to j, the self-attention coefficients $\boldsymbol{\alpha}_{c,ij}^{(l)}(\boldsymbol{q},\boldsymbol{k})$ are calculated by multihead dot product attention. Then, multiply the self-attention coefficients with the projected features of node j to obtain dynamically weighted features. The residual connection is used by adding the input to the output of this GT block to obtain updated features $\boldsymbol{attention}_i^{(l+1)}$. Finally, an activation function LeakyRelu and a Dropout layer are used to normalize the results and reduce overfitting issues. The self-attention coefficients $\boldsymbol{\alpha}_{c,ij}^{(l)}(\boldsymbol{q},\boldsymbol{k})$, updated features $\boldsymbol{attention}_i^{(l+1)}$, and final results of each GT block $\boldsymbol{H}^{(l+1)}$ are calculated as follows:

$$\begin{aligned} \boldsymbol{\alpha}_{c,ij}^{(l)}(\boldsymbol{q},\boldsymbol{k}) &= \mathrm{softmax}\left(\frac{\boldsymbol{q}_{c,i}^{(l)}\boldsymbol{k}_{c,j}^{(l)\,T}}{\sqrt{d}}\right) \\ &= \frac{\exp\left(\frac{\boldsymbol{q}_{c,i}^{(l)}\boldsymbol{k}_{c,j}^{(l)\,T}}{\sqrt{d}}\right)}{\sum_{\boldsymbol{A}_{ij}\neq 0}\exp\left(\frac{\boldsymbol{q}_{c,i}^{(l)}\boldsymbol{k}_{c,j}^{(l)\,T}}{\sqrt{d}}\right)} \end{aligned} \tag{3}$$

where $\boldsymbol{q} \in \mathbb{R}^d$ and $\boldsymbol{k} \in \mathbb{R}^d$ denote the ***query*** vector and ***key*** vector of multihead dot product attention, respectively; $\boldsymbol{\alpha}_{c,ij}$ represents the weight coefficient of node j to node i in the cth multihead self-attention; d is the size of channel number, c is the number of multihead self-attention, L is the number of GT blocks, and $l \in \{1, 2, \ldots, L\}$

$$\begin{aligned} \boldsymbol{attention}_i^{(l+1)}(\boldsymbol{q},\boldsymbol{k},\boldsymbol{v}) &= \boldsymbol{W}_{c,1}^{(l)}\boldsymbol{h}_i^{(l)} + \Big\|_{c=1}^{C}\sum_{\boldsymbol{A}_{i,j}\neq 0}\boldsymbol{\alpha}_{c,ij}^{(l)}\boldsymbol{W}_{c,2}^{(l)}\left(\boldsymbol{v}_{c,j}^{(l)}\right) \\ \boldsymbol{h}_i^{(l+1)} &= \mathrm{Dropout}\left(\sigma\left(\boldsymbol{attention}_i^{(l+1)}\right)\right) \end{aligned} \tag{4}$$

where $\boldsymbol{h}_i^{(l+1)}$ is the ith element of $\boldsymbol{H}^{(l+1)}$, $\boldsymbol{v} \in \mathbb{R}^d$ represents the ***value*** vector of multihead dot product attention, $\sigma(\cdot)$ is the LeakyRelu function, and $\boldsymbol{W}$ is the weight matrix.

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Finally, by repeating L GT blocks, the final topology structure features are denoted as $\boldsymbol{H}^{(L+1)}$. Superpixel-level structure features can be transformed into features ***Spf*** at pixel space by multiplying the association matrix with the output features of GT module. In the dual-branch GA module for multitask meta-optimization and three-branch GA module for HCD task, it can be, respectively, calculated as follows:

$$\begin{aligned}\boldsymbol{Spf} &= \boldsymbol{Q}_0 \cdot \boldsymbol{H}_0^{(L+1)} \\ \boldsymbol{Spf} &= \boldsymbol{Z}_1 - \boldsymbol{Z}_2 = \left(\boldsymbol{Q}_1 \cdot \boldsymbol{H}_1^{(L+1)}\right) - \left(\boldsymbol{Q}_2 \cdot \boldsymbol{H}_2^{(L+1)}\right)\end{aligned} \tag{5}$$

where $\boldsymbol{Q}_1$ and $\boldsymbol{Q}_2$ are the association matrix for bitemporal HSIs, $\boldsymbol{H}_1^{(L+1)}$ and $\boldsymbol{H}_1^{(L+1)}$ are the superpixel-level structure feature matrix for bitemporal HSIs obtained by L GT blocks, $\boldsymbol{Q}_0$ is the association matrix of single HSI, and $\boldsymbol{H}_0^{(L+1)}$ is the superpixel-level structure features of single HSI.

3) Graph-Enhanced Feature Embedding: PMA can fully extract the pixel-level spatial–spectral features but ignores the significant spatial features corresponding to the topology structures of objects. While the GT module can flexibly process the irregular structures but overlooks the abundant spectral information of HSIs and the pixel-level differences within superpixels. In order to comprehensively extract the valuable features of HSIs, we integrate the topological structural features ***Spf*** with spatial–spectral features ***Pf*** to enhance the pixel-level feature representation of images, thereby obtaining graph-enhanced pixel-level spatial–spectral features $\boldsymbol{P}$

$$\boldsymbol{P} = \mathrm{FC}(\mathrm{Concat}(\boldsymbol{Spf}, \quad \boldsymbol{Pf})) \tag{6}$$

where ***Spf*** is the topology structure features, ***Pf*** is the pixel level spatial–spectral features obtained by PMA module, $FC(\cdot)$ denotes the fully connected layers, and $Concat(\cdot)$ is the concatenation operation.

C. Multitask Meta-Optimization

In this section, the model based on dual-branch GA module is pretrained by meta-learning strategy to learn the general knowledge suitable for across tasks, thereby enabling the pretrained model to learn the ability of efficiently handling binary classification tasks. The hyperspectral classification datasets are easier to label than multiple HCD datasets and usually have more class information. We propose to use the rich class information in HSIs to construct multiple binary classification tasks and learn the correlation of these tasks through meta-optimization to optimize the model based on the dual-branch GA module. The model obtained after multitask meta-optimization can be quickly adapted to distinguish different classes, including different change classes and unchanged classes in CD tasks.

In practice, we utilize two hyperspectral datasets to construct T binary classification tasks $\{\tau_1, \tau_2, \ldots, \tau_i, \ldots, \tau_T\}$. With the rich class information of hyperspectral classification dataset, it is used to make the model fully learn the ability of identifying land covers in HSIs. Since the main goal of CMGA is accurately identifying change classes with temporal information, an HCD dataset is used as a supplement to construct similar tasks to the hyperspectral classification dataset, thereby making the optimized model more suitable for adapting to HCD tasks. All the samples with the same label are correspondingly divided into support datasets $\chi^{\mathrm{sup}} = \{\chi_{\tau_1}^{\mathrm{sup}}, \chi_{\tau_2}^{\mathrm{sup}}, \ldots, \chi_{\tau_i}^{\mathrm{sup}}, \ldots, \chi_{\tau_T}^{\mathrm{sup}}\}$ and query datasets $\chi^{\mathrm{que}} = \{\chi_{\tau_1}^{\mathrm{que}}, \chi_{\tau_2}^{\mathrm{que}}, \ldots, \chi_{\tau_i}^{\mathrm{que}}, \ldots, \chi_{\tau_T}^{\mathrm{que}}\}$. Each pair of samples in support datasets and query datasets with same label comprises one binary classification task. For task τ_i, the support data for within-task training and query data for within-task testing can be denoted as $\chi_{\tau_i}^{\mathrm{sup}}$ and $\chi_{\tau_i}^{\mathrm{que}}$, respectively. During within-task training, the model learns how to make appropriate predictions based on $\chi_{\tau_i}^{\mathrm{sup}}$ and update the parameters. In the within-task testing phase of meta-optimization, the model f_ϕ evaluates its performance on the current task using $\chi_{\tau_i}^{\mathrm{que}}$ and calculates the corresponding testing loss $\mathrm{loss}_{\chi_{\tau_i}^{\mathrm{que}}}^{\mathrm{tra}}$. By averaging the testing loss of all tasks, the model can balance the contributions of different tasks and thus learn the general meta-knowledge across tasks. The mathematical expressions of multitask meta-optimization are as follows:

The within-task training is used to update the randomly initialized parameters ϕ by gradient descent algorithm for each task

$$\phi_i^k = \phi_i^{k-1} - \varepsilon \nabla_{\phi_i^{k-1}} \mathrm{loss}_{\chi_{\tau_i}^{\mathrm{sup}}}^{\mathrm{tra}}\left(\phi_i^{k-1}\right) \tag{7}$$

where $l_{\chi_{\tau_i}^{\mathrm{sup}}}^{\mathrm{tra}}$ is the cross-entropy loss function for task τ_i on $\chi_{\tau_i}^{\mathrm{sup}}$, ϕ_i^k represents the updated parameters of the kth within-task training, $\phi_i^0 = \phi$, and ε is the learning rate of within-task training for each task.

By minimizing the average testing loss obtained from all tasks, we can obtain the optimal model parameters $\widehat{\phi}$

$$\widehat{\phi} = \arg\min_{\phi} \sum_{i}^{T} \mathrm{loss}_{\chi_{\tau_i}^{\mathrm{que}}}^{\mathrm{tra}}(\phi_i). \tag{8}$$

According to the chain rule [48], based on all the data for within-task testing, the second-order gradient is used to update parameters

$$\begin{aligned}\widehat{\phi} &\leftarrow \phi - \eta \cdot g_{\phi}^{\mathrm{second}} \\ &= \phi - \eta \cdot \sum_{i}^{T} \nabla_{\phi} \mathrm{loss}_{\chi_{\tau_i}^{\mathrm{que}}}^{\mathrm{tra}}(\phi_i) \\ &= \phi - \eta \cdot \sum_{i}^{T} \nabla_{\phi_i} \mathrm{loss}_{\chi_{\tau_i}^{\mathrm{que}}}^{\mathrm{tra}}(\phi_i) \cdot \nabla_{\phi}(\phi_i) \\ &= \phi - \eta \cdot \sum_{i}^{T} \nabla_{\phi_i} \mathrm{loss}_{\chi_{\tau_i}^{\mathrm{que}}}^{\mathrm{tra}}(\phi_i) \cdot \left(\boldsymbol{I} - \varepsilon \boldsymbol{\Omega}_{\phi}\left(\mathrm{loss}_{\chi_{\tau_i}^{\mathrm{sup}}}^{\mathrm{tra}}(\phi_i)\right)\right)\end{aligned} \tag{9}$$

where $\boldsymbol{\Omega}_{\phi}(\mathrm{loss}^{\mathrm{tra}})$ is the Hessian matrix with $(\boldsymbol{\Omega}_{\phi}(\mathrm{loss}^{\mathrm{tra}}))_{ij} = \partial^2 \mathrm{loss}^{\mathrm{tra}} / \partial \phi_i \phi_j$ and η is the learning rate of within-task testing.

D. Meta-Knowledge-Guided Change Detection

In this section, we describe how to transfer the meta-knowledge to guide HBCD and HMCD tasks. First, the constructed graph structures $\boldsymbol{G}_1 = (\boldsymbol{A}_1, \boldsymbol{H}_1)$ and $\boldsymbol{G}_2 = (\boldsymbol{A}_2, \boldsymbol{H}_2)$ of bitemporal HSIs are served as inputs to a Siamese network formed by the pretrained GT module that is composed

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of several GT blocks. Then, the difference between bitemporal HSIs is calculated and served as input to the pretrained PMA module, aiming to extract deep pixel-level change features of spatial–spectral information. The output features of GT and PMA modules are concatenated to obtain graph-enhanced pixel-level spatial–spectral features. In order to accelerate gradient descent and fully utilize spatial–spectral information of diverse HSIs, the parameters of GT module are frozen and the parameters of low-parametric PMA module are continued for training to comprehensively extract spatial–spectral features of objects. Finally, the pretrained prediction head composed of fully connection layers and a softmax activation function are continued for training to obtain CD results. Besides, we propose a cosine similarity loss to amplify the feature differences obtained by the Siamese structure for effectively detecting changed areas. By designing a scoring system for progressive detection strategy, HBCD and HMCD tasks are unified into the identical classification structure of the meta-knowledge-guided CD process. Therefore, the pretrained dual-branch GA module is extended to three-branch and is quickly adapted to CD tasks. The detailed overall framework is shown in Fig. 2.

1) Learning Strategy for HBCD and HMCD: Given the HCD dataset with δ change classes and one unchanged class, a progressive detection strategy is used to construct δ binary classifiers. A few labeled data $\{\aleph_1^{\text{tra}}, \aleph_2^{\text{tra}}, \ldots, \aleph_i^{\text{tra}}, \ldots, \aleph_\delta^{\text{tra}}\}$ are randomly selected to adapt the pretraining model for CD tasks, and the others $\{\aleph_1^{\text{tes}}, \aleph_2^{\text{tes}}, \ldots, \aleph_i^{\text{tes}}, \ldots, \aleph_\delta^{\text{tes}}\}$ are used for prediction. Each binary classifier identifies one change class from the other classes including the unchanged class. For each pixel, the scoring system stores the predicted classes of all binary classifiers and determines the final class based on the highest score in all the predicted classes. If the scores for all predicted classes are zero, the pixel is assigned to the unchanged class. The learning strategy can be defined as follows:

$$\widehat{\phi}_i^{\,j} = \widehat{\phi}_i^{\,j-1} - \mu \nabla_{\widehat{\phi}_i^{\,j-1}} \text{Loss}_{\aleph_i^{\text{tra}}}\left(\widehat{\phi}_i^{\,j-1} \middle| \widehat{\theta}\right) \tag{10}$$

where $\aleph_i^{\text{tra}}$ represents the data used to adapt the model for identifying class i, μ is the learning rate for every binary classifier, j is the current number of epochs, and $\text{Loss}(\widehat{\phi}|\widehat{\theta})$ denotes freezing the parameters $\widehat{\theta}$ of GT module and only updating the other parameters of the model.

Then, the optimal parameters $\widehat{\phi}_i^{\,*}$ for each binary classifier are obtained by minimizing the loss function

$$\widehat{\phi}_i^{\,*} = \arg\min_{\widehat{\phi}_i} \text{Loss}_{\aleph_i^{\text{tra}}}\left(\widehat{\phi}_i \middle| \widehat{\theta}\right) \tag{11}$$

where the initial parameters of each binary classifier are represented as $\phi_i^0 = \widehat{\phi}$, which is obtained from multitask meta-optimization.

2) Loss Function: In addition to utilize the cross-entropy loss function $\text{Loss}_{\text{ent}}(\hat{Y}, Y)$, a cosine similarity loss function $\text{Loss}_{\cos}(\mathbf{Z}_1, \mathbf{Z}_2)$ is designed for CD tasks. The cosine similarity loss function calculates the mean cosine similarity between pixel-level topology structure feature vectors of all corresponding pixels, representing the overall difference in features between the bitemporal HSIs. A higher value indicates a less pronounced distinction between pixel-level topology structure features. Therefore, by reducing the value of cosine similarity loss function, the differences between bitemporal HSIs are amplified, making the extracted features more conducive to identify changed areas. The cosine similarity loss function can be represented as

$$\text{Loss}_{\cos}(\mathbf{Z}_1, \mathbf{Z}_2) = \frac{1}{N}\sum_{i=1}^{N} \frac{(\mathbf{Z}_1)_i \cdot (\mathbf{Z}_2)_i}{\|(\mathbf{Z}_1)_i\| \|(\mathbf{Z}_2)_i\|} \tag{12}$$

where $\mathbf{Z}_1$ and $\mathbf{Z}_2$ denote the superpixel-level structure features of bitemporal HSIs and N is the total number of pixels.

The total loss function $Loss$ for CD tasks is defined as follows:

$$\text{Loss}_{\text{ent}}\left(\hat{\boldsymbol{Y}}, \boldsymbol{Y}\right) = -\frac{1}{S}\sum_{s=1}^{S}\left\{Y_s \log\left(\hat{Y}_s\right) + (1-Y_s)\log\left(1-\hat{Y}_s\right)\right\}$$
$$\text{Loss} = \text{Loss}_{\text{ent}}\left(\hat{\boldsymbol{Y}}, \boldsymbol{Y}\right) + \lambda \text{Loss}_{\cos}(\mathbf{Z}_1, \mathbf{Z}_2) \tag{13}$$

where $\text{Loss}_{\text{ent}}\left(\hat{\boldsymbol{Y}}, \boldsymbol{Y}\right)$ is used to calculate the difference between the truth labels $\boldsymbol{Y}$ and predict labels $\hat{\boldsymbol{Y}}$; S denotes the number of training samples; and λ is the weight coefficient for cosine similarity loss, which is set to 0.3 through a series of experiments.

Finally, the overall learning process for CMGA can be summarized as Algorithm 1.

IV. Experiments

A. Experimental Datasets

In this work, one hyperspectral classification dataset and five CD datasets [15], [60], [61], [62], [63] are used, namely, Indian Pines, Santa Barbara, China, USA, Benton, and River. The Indian Pines and Santa Barbara are used as the meta-training datasets, while the others are used as the meta-testing datasets. The details of each dataset are given here.

1) Indian Pines Dataset[1]*:* The HSI was captured using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in Northwest India. The dataset includes 200 spectral bands and measures 145 × 145 pixels with a spatial resolution of 20 m/pixel. It includes 16 classes, as shown in Table I.

2) Santa Barbara Dataset[2]*:* The two HSIs were obtained in 2013 and 2014, near Santa Barbara, CA, USA. They were obtained by the AVIRIS sensor, making up of mountainous and farmlands. With the size of 984 × 740 pixels and 224 bands, the ground-truth image contains 52 134 changed pixels and 80 148 unchanged pixels, and the rest are undetermined pixels.

3) China Dataset[3]*:* The two HSIs were obtained on May 3, 2006 and April 23, 2007, near the city of Yancheng, Jiangsu, China. They were obtained by the Earth Observing-1 (EO-1) Hyperion hyperspectral sensor, covering an area of farmlands with the size of 420 × 140 pixels and 155 bands. There are three change classes in this dataset, as shown in Table II.

[1] https://rslab.ut.ac.ir/data

[2] https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

[3] https://github.com/meiqihu/BCG-Net

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Algorithm 1 Overall Learning Process for CMGA

Input: $\boldsymbol{X}_{\text{tra}}$-input HSI for multi-task meta optimization process; $\boldsymbol{X}_{\text{tes}}$-input HSI for meta-knowledge guided change detection process; Overall pipeline f_ϕ with PMA module, GT module parameterized by θ, and fully connection layers

Output: Change detection results $\hat{\boldsymbol{Y}}$

// Data Pre-processing

Construct superpixel graph structures $\boldsymbol{G}_0$ for multi-task meta optimization process and $\boldsymbol{G}_1, \boldsymbol{G}_2$ for meta-knowledge guided change detection process

Construct T tasks $\tau = \{\tau_1, \tau_2, \ldots, \tau_i, \ldots, \tau_T\}$ with support datasets $\chi^{\text{sup}} = \{\chi^{\text{sup}}_{\tau_1}, \chi^{\text{sup}}_{\tau_2}, \ldots, \chi^{\text{sup}}_{\tau_i}, \ldots, \chi^{\text{sup}}_{\tau_T}\}$ and query datasets $\chi^{\text{que}} = \{\chi^{\text{que}}_{\tau_1}, \chi^{\text{que}}_{\tau_2}, \ldots, \chi^{\text{que}}_{\tau_i}, \ldots, \chi^{\text{que}}_{\tau_T}\}$ for multi-task meta optimization process

Based on δ change classes, sample training datasets $\aleph^{\text{tra}} = \{\aleph^{\text{tra}}_1, \aleph^{\text{tra}}_2, \ldots, \aleph^{\text{tra}}_i, \ldots, \aleph^{\text{tra}}_\delta\}$ for meta-knowledge guided change detection process

// Multi-task Meta Optimization

Initialize parameters ϕ of the overall pipeline

for i in 1: T **do**

  // Within-task training

  Sample $\tau_i \in \tau$ with support data $\chi^{\text{sup}}_{\tau_i}$ and query data $\chi^{\text{que}}_{\tau_i}$

  for k in 1 : epoch **do**

    $\phi_i \leftarrow \phi - \varepsilon \nabla_{\phi_i} loss^{\text{tra}}_{\chi^{\text{sup}}_{\tau_i}}(f_{\phi_i})$

  end

  // Within-task testing

  Update ϕ by $\widehat{\phi} = \arg\min_{\phi} \sum_{i}^{T} loss^{\text{tra}}_{\chi^{\text{que}}_{\tau_i}}(\phi_i)$

end

Obtain the updated pipeline $f_{\widehat{\phi}}$

// Meta-knowledge Guided Change Detection

Froze parameters $\widehat{\theta}$ of GT module

for i in 1 : δ **do**

  Update $\widehat{\phi}$ by Equation (8) on $\aleph^{\text{tra}}_i$

  Obtain the optimal pipeline $f_{\widehat{\phi}^*|\widehat{\theta}}$

  Predict changes on class i by $\hat{\boldsymbol{Y}}_i = f_{\widehat{\phi}^*|\widehat{\theta}}(\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{X}_{\text{tes}})$

end

Obtain multiclass change results by $\hat{\boldsymbol{Y}} = Max\left(\sum_{i=1}^{\delta} \hat{\boldsymbol{Y}}_i\right)$

return $\hat{\boldsymbol{Y}}$

4) USA Dataset[4]*:* The two HSIs were obtained on May 1, 2004 and May 8, 2007, in Hermiston city, Oregon, USA, by the EO-1 Hyperion sensor. They cover an area of irrigated agriculture with the size of 307 × 241 pixels and 156 bands. This dataset includes six change classes, as shown in Table II.

[4]https://github.com/meiqihu/BCG-Net

TABLE I

DETAILS OF CLASS INFORMATION FOR THE INDIAN PINES DATASET

Class	Class Name	Sample Number
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
	TOTAL	10249

TABLE II

DETAILS OF CHANGE INFORMATION FOR FOUR DATASETS

Class	Sample Number			
	China	USA	Benton	River
Change-1	6559	1850	1034	-
Change-2	11613	2905	1048	-
Change-3	105	5896	5111	-
Change-4	-	1524	1261	-
Change-5	-	2744	479	-
Change-6	-	1808	988	-
Changed	18277	16727	9921	9698
Unchanged	40467	57260	30579	101885

TABLE III

CD RESULTS OF ABLATION STUDY ON LOSS FUNCTION

Case	Metric	China	USA	Benton	River
④	OA	97.60	95.87	96.35	96.12
	Kappa	94.45	88.63	90.01	70.19
	F1	97.22	94.31	95.01	85.03
	Training time/s	1.38	5.51	1.96	6.14
⑤	OA	**97.67**	**96.13**	**96.53**	96.12
	Kappa	**94.60**	**89.27**	**90.45**	70.19
	F1	**97.30**	**94.63**	**95.22**	85.03
	Training time/s	1.37	5.52	1.97	6.19

5) Benton Dataset[5]*:* The two HSIs were obtained on May 1, 2004 and May 8, 2007, in Benton country, Oregon, USA. They were obtained by the EO-1 Hyperion sensor, covering an area of irrigated agriculture with the size of 180 × 225 pixels and 159 bands. There are six change classes, as shown in Table II.

6) River Dataset[6]*:* The two HSIs were obtained on May 3, 2013 and December 31, 2013, in Jiangsu province, China. They were also obtained by the EO-1 Hyperion sensor, mainly covering an area of river with the size of 463 × 241 pixels and 198 bands. The ground-truth image contains 9698 changed pixels and 101 885 unchanged pixels.

For HBCD tasks, the ground-truth image consists of the changed regions in white, the unchanged regions in black,

[5]https://github.com/SicongLiuRS/Hyperspectral-Change-DetectionDataset-Irrigated-Agricultural-Area

[6]https://crabwq.github.io

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TABLE IV
PERFORMANCE OF MULTICLASS CHANGE DETECTION ON THREE DATASETS (TIME UNIT: SECONDS)

Dataset	Method	BCG-Net	ML-EDAN	MSDFFN	D^{2}AGCN	MGCN	GTransCD	CMGA
China	Unchanged	96.60	98.13	97.85	95.17	97.80	90.44	98.12
	Change-1	87.68	90.63	91.25	91.85	89.99	99.72	94.76
	Change-2	98.57	94.04	95.61	96.35	92.83	99.74	97.56
	Change-3	16.67	00.00	00.00	21.15	00.00	4.81	46.43
	OA	95.59	96.42	96.64	94.90	95.87	93.16	**97.58**
	Kappa	90.69	92.52	92.94	89.41	91.39	86.53	**94.89**
	Precision	59.90	70.70	71.18	76.13	70.16	73.68	**84.22**
	F1	58.96	71.46	71.61	77.58	70.89	69.98	**80.70**
	Training time	378.01	64.46	93.61	5409.91	211.37	81.44	**5.89**
	Testing time	0.03	4.06	8.07	4.52	0.05	**0.01**	0.04
USA	Unchanged	96.41	91.98	93.71	97.42	91.34	98.31	93.83
	Change-1	55.13	14.53	75.15	20.78	51.66	44.40	65.42
	Change-2	96.51	20.37	46.19	32.38	50.62	37.36	69.02
	Change-3	91.03	66.64	79.17	87.49	75.02	81.82	90.81
	Change-4	45.34	32.14	47.26	52.64	41.78	62.66	47.99
	Change-5	63.42	69.59	62.26	75.46	70.58	62.55	81.31
	Change-6	50.00	50.00	72.36	28.71	55.68	3.51	59.88
	OA	83.61	84.95	88.65	88.74	87.35	88.88	**90.77**
	Kappa	58.19	58.28	68.59	68.59	63.12	70.15	**74.28**
	Precision	62.23	49.32	68.01	56.41	62.38	55.80	**72.61**
	F1	41.69	40.40	58.17	60.26	49.49	55.33	**65.01**
	Training time	321.52	81.66	236.59	1958.52	105.16	97.99	**37.74**
	Testing time	0.03	5.04	20.70	1.65	0.03	**0.01**	0.39
Benton	Unchanged	-	96.32	94.49	97.08	95.37	98.65	97.45
	Change-1	-	68.50	91.02	53.60	95.89	11.82	71.64
	Change-2	-	93.77	92.23	89.71	93.18	97.15	94.77
	Change-3	-	56.48	73.52	79.49	77.88	82.20	84.81
	Change-4	-	18.42	94.57	91.18	84.69	17.02	91.74
	Change-5	-	8.77	71.19	86.16	91.15	70.60	69.36
	Change-6	-	56.71	91.57	94.74	80.14	91.81	91.19
	OA	-	91.37	93.03	94.08	93.91	93.88	**95.10**
	Kappa	-	79.15	81.72	85.44	84.47	84.31	**88.00**
	Precision	-	57.00	86.94	84.56	**88.33**	67.03	85.85
	F1	-	57.91	68.15	83.30	75.04	69.76	**84.37**
	Training time	-	54.57	134.75	883.69	47.61	259.75	**13.91**
	Testing time	-	2.80	11.04	0.76	0.02	**0.01**	0.12

and the undetermined regions in gray. For HMCD tasks, the ground-truth image consists of the changed regions in different colors and the unchanged regions in black.

B. Experimental Setting

1) Meta-Optimization Setting: We utilize the optimizer with the learning rate of 0.0001 across 500 epochs to optimize the pretraining model. A total of 50% labeled samples are randomly selected from Indian Pines and Santa Barbara datasets to construct support datasets, and the others are used as query datasets. The segmentation scale for both Indian Pines and Santa Barbara datasets is empirically set to 25 [26].

2) Change Detection Setting: The Adam optimizer is also used with the learning rate of 0.01 across 70 epochs to optimize the CD model. In the experiments, we randomly select a total of 0.5% labeled samples for training to ensure fairness. We experimentally set the best segmentation scale for the China, USA, Benton, and River datasets as 50, 5, 5, and 10, respectively. According to experiments, the values of both L and c are set to 2. The numbers of hidden layers for the first and the second GT block are set as 128 and 100, respectively. The numbers of hidden layers for conv1 $\times$ 1, conv3 $\times$ 3, and conv5 $\times$ 5 in PMA are set as 128. Also, all the experiments are conducted on NVIDIA GeForce RTX 3090 based on the PyTorch framework.

The overall accuracy (OA), Kappa coefficient, precision, $F1$ score, training time, testing time, and trainable parameters are calculated for quantitative evaluation of both HBCD and HMCD tasks. To illustrate the effectiveness of CMGA, six classical and state-of-the-art methods were conducted for comparison. The ML-EDAN [16], MSDFFN [17], D^{2}AGCN [33], MGCN [35], and GTransCD [64] are selected for comparison on both HBCD and HMCD tasks. The above four methods both utilize a fully connection layer with two channels to detect changed and unchanged areas in a binary classification paradigm. By setting the channel number of the final fully connection layer as the number of classes, the four methods are used to distinguish various change classes. The BCG-Net [6] is utilized for comparison on HMCD tasks, and the SST-Former [21] is used on HBCD tasks.

C. Ablation Study

The GT module, PMA module, and cosine similarity loss function play significant roles in our network. In this section, the effectiveness of them is demonstrated by the ablation study.

1) Ablation Study on GT Module and PMA Module: The basic architecture is based on the pretrained model. With the

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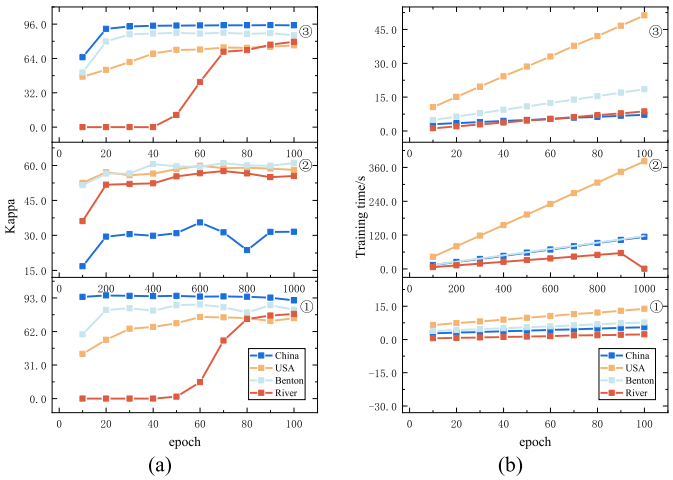


Fig. 3. Effect of different modules on four datasets. (a) Kappa. (b) Training time.

increase of epoch, the effect of GT module and PMA module is illustrated in Fig. 3. The results are provided via Kappa and training time. Three cases are compared: 1) using only PMA module; 2) using only GT module; and 3) using both PMA and GT modules.

As shown in Fig. 3, compared to using only PMA or GT, using both two modules can obtain the highest value of Kappa during the training process. The performance on the best epoch of combining PMA module and GT module is better than that of only using the PMA module and GT module. Compared with using only GT module, graph-enhanced pixel-level feature embedding can use the topology structure features to enhance the pixel-level feature representation with superior computational complexity.

2) Ablation Study on Loss Function: We further conduct experiments to analyze the effect of total loss function. The experiments are conducted on four datasets for HBCD tasks. Two cases are compared: 1) using only cross-entropy loss function and 2) using total loss function with cosine similarity loss and cross-entropy loss function.

The experimental results are provided via OA, Kappa, $F1$, and training time, as shown in Table III. Without increasing the computational complexity, the cosine similarity loss function enhances the CD results for most datasets. For example, the OA, Kappa, and $F1$ for HBCD task on USA dataset are increased by 0.26%, 0.64%, and 0.32%, respectively. Also, the training time remains almost unchanged. Therefore, the cosine similarity loss function is helpful to amplify the differences between bitemporal HSIs.

D. Change Detection Results

In this section, the experiments are conducted five times, and the third-ranked value for each metric is used as the final evaluation. Table IV shows the HMCD results on three datasets, and Table V shows the HBCD results on four datasets, where the highest value is in bold and the second highest value is underlined. The multiclass and binary change maps are shown from Figs. 4 to 6 and Figs. 7 to 10, respectively.

1) Experiment results of HMCD: As shown in Table I, the results of the CMGA achieve the state of the art for HMCD on three datasets, illustrating the ability of addressing new classes for meta-knowledge-guided CD. For example, compared with the second highest value, the OA, Kappa, precision, and $F1$ for the HMCD task on the USA dataset are increased by 1.89%, 4.13%, 4.6%, and 4.75%, respectively. Also, compared with D^{2}AGCN, MGCN, and GTransCD, the training time of CMGA on USA dataset is reduced by 1920.78, 67.42, and 60.25 s, respectively. When dealing with subtle change classes, CMGA has a distinct advantage, such as China dataset, CMGA can increase the accuracy of Change-3 by 25.28%. As shown in Figs. 4–6, the comparison methods usually mistakenly classify a change class as others, but CMGA can accurately identify various change classes. Specifically, the comparison methods fail to identify the change class in the middle of China dataset, and CMGA can clearly display this change class on multiclass change map.

2) Experiment results of HBCD: As shown in Table V, CMGA performs optimally on the China, USA, Benton, and River datasets, with training times all under 10 s, demonstrating high generalization and efficiency. In detail, compared with the second highest values obtained by MGCN, the OA, Kappa, precision, and $F1$ for HBCD task on th Benton dataset are increased by 1.73%, 4.39%, 2.05%, and 2.19%, respectively. In comparison of GNN-based methods D^{2}AGCN, MGCN, and GTransCD, the training time of CMGA on the Benton dataset is reduced by 825.87, 9.96, and 206.51 s, respectively. In addition, as shown in Figs. 7–10, compared with other CD methods, CMGA can better extract changed regions that are usually mistakenly classified as unchanged regions, such as the changed regions in the bottom of the USA dataset, and the subtle changed regions in the middle part of the China dataset. What is more, the trainable parameters number of CMGA is only 0.11 M, which is superior to most of the methods.

E. Parameter Analysis

In this section, we first illustrate the influence of superpixel segmentation scale on CMGA. Then, the performance of CMGA with different values of L and c is discussed. Finally, the influence of training samples is evaluated. HMCD results are illustrated on China, USA, and Benton datasets, and HBCD results are illustrated on the River dataset for all experiments in this section.

1) Effect of Superpixel Segmentation Scale: The OA, Kappa, and training time are calculated on four datasets to analyze the influence of superpixel segmentation scale, as shown in Fig. 11. For the China dataset with many regular objects, the training time is the first metric we considered; then, according to values of other metrics, 50 is set as the best segmentation scale. To better detect many small changed regions of USA dataset, the smallest segmentation scale 5 is used in CMGA. According to the best values of OA and Kappa, the segmentation scale of Benton dataset used in CMGA is 5. For the River dataset, as the segmentation scale increases, the values of OA and Kappa gradually decrease, and the training time suddenly decreases. Therefore, comprehensively considering the three

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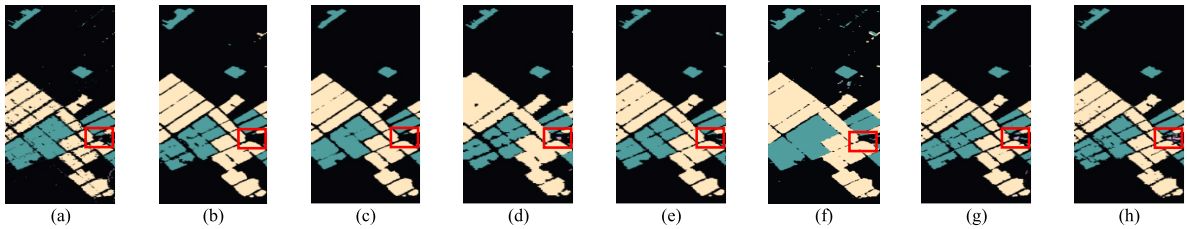


Fig. 4. Multiclass change maps of different methods on the China dataset. (a) BCG-Net. (b) ML-EDAN. (c) MSDFFN. (d) D^{2}AGCN. (e) MGCN. (f) GTransCD. (g) CMGA. (h) Ground-truth image.

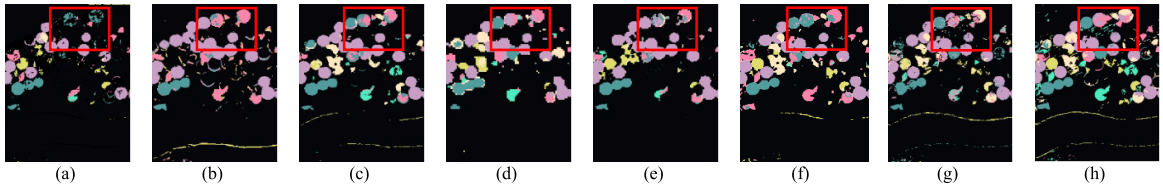


Fig. 5. Multiclass change maps of different methods on the USA dataset. (a) BCG-Net. (b) ML-EDAN. (c) MSDFFN. (d) D^{2}AGCN. (e) MGCN. (f) GTransCD. (g) CMGA. (h) Ground-truth image.

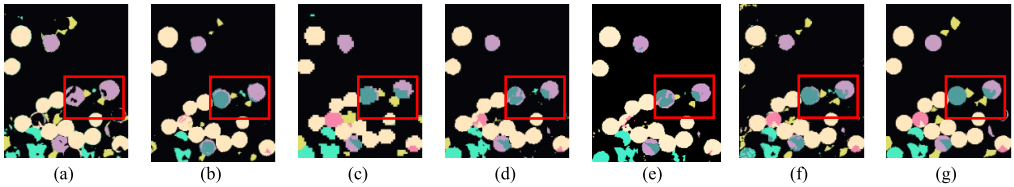


Fig. 6. Multiclass change maps of different methods on the Benton dataset. (a) ML-EDAN. (b) MSDFFN. (c) D^{2}AGCN. (d) MGCN. (e) GTransCD. (f) CMGA. (g) Ground-truth image.

TABLE V

PERFORMANCE OF BINARY CHANGE DETECTION ON FOUR DATASETS (TIME UNIT: SECONDS)

Dataset	Metric	ML-EDAN	MSDFFN	SST-Former	D^{2}AGCN	MGCN	GTransCD	CMGA
China	OA	95.40	96.50	96.23	95.27	96.62	94.16	**97.67**
	Kappa	89.29	91.82	91.34	89.03	92.19	86.96	**95.60**
	Precision	94.57	96.00	0.9501	0.9475	95.70	95.43	**96.98**
	F1	94.64	95.91	95.67	94.51	96.09	93.46	**97.30**
	Training time	65.51	106.74	114.27	5419.82	55.24	54.15	**1.37**
	Testing time	4.06	7.77	5.69	4.51	0.09	**0.01**	0.04
USA	OA	92.25	94.39	94.61	91.34	95.11	95.04	**96.13**
	Kappa	78.15	83.27	84.70	73.70	86.01	85.97	**89.27**
	Precision	88.51	**93.60**	91.84	84.78	92.93	93.16	93.34
	F1	89.07	91.63	92.35	86.83	93.01	92.93	**94.63**
	Training time	98.24	124.57	59.42	1855.72	48.40	99.86	**5.52**
	Testing time	5.13	10.17	7.06	1.57	0.06	**0.01**	0.05
Benton	OA	93.30	93.65	93.56	94.70	94.80	94.35	**96.53**
	Kappa	80.78	81.81	83.19	85.27	86.06	84.37	**90.45**
	Precision	93.50	93.98	90.22	91.46	92.66	91.22	**96.03**
	F1	90.37	90.88	91.59	92.63	93.03	92.18	**95.22**
	Training time	56.83	71.92	54.39	827.84	11.93	208.48	**1.97**
	Testing time	2.96	5.41	4.26	0.71	0.02	**0.01**	0.03
River	OA	94.97	94.92	93.03	94.04	94.29	95.47	**96.12**
	Kappa	65.00	61.28	65.57	60.28	61.37	67.94	**70.19**
	Precision	86.45	89.68	77.29	78.47	83.24	80.22	**95.99**
	F1	82.48	80.55	82.64	80.13	80.67	83.95	**85.03**
	Training time	117.51	194.62	157.74	3868.87	59.76	64.82	**6.19**
	Testing time	8.05	15.14	12.44	3.24	0.05	**0.01**	0.08
Trainable Parameters(M)		112.29	26.13	2.49	2.15	0.10	146.11	0.11

metrics, the best segmentation scale is set as 10 for the River dataset.

2) Effect of L and c: The Kappa and training time are used on four datasets to analyze the performance of CMGA with different GT blocks and multihead attentions. First, as shown in Fig. 12, CMGA performs well when L is 2 and 4. Considering that training time increases with the increase in values of L, the value of 2 is determined as the best value of L. Based on $L = 2$, as shown in Fig. 13, when c is 2, CMGA performs well on four datasets. Therefore,

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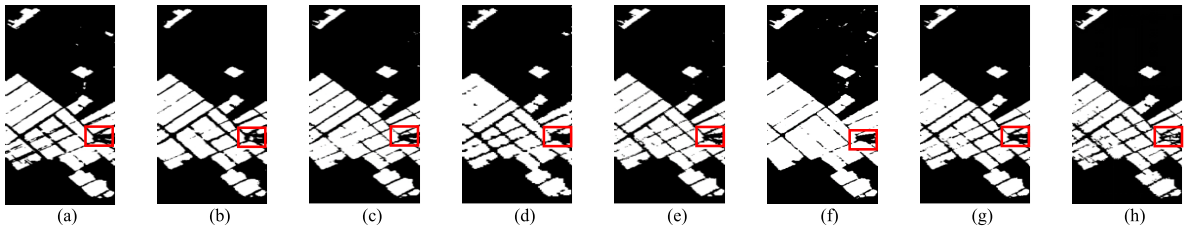


Fig. 7. Binary change maps of different methods on the China dataset. (a) ML-EDAN. (b) MSDFFN. (c) SST-Former. (d) D^{2}AGCN. (e) MGCN. (f) GTransCD. (g) CMGA. (h) Ground-truth image.

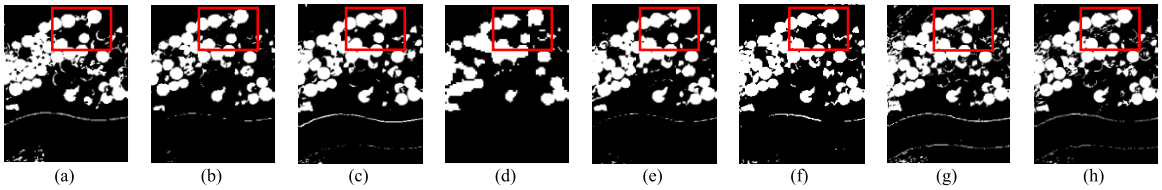


Fig. 8. Binary change maps of different methods on the USA dataset. (a) ML-EDAN. (b) MSDFFN. (c) SST-Former. (d) D^{2}AGCN. (e) MGCN. (f) GTransCD. (g) CMGA. (h) Ground-truth image.

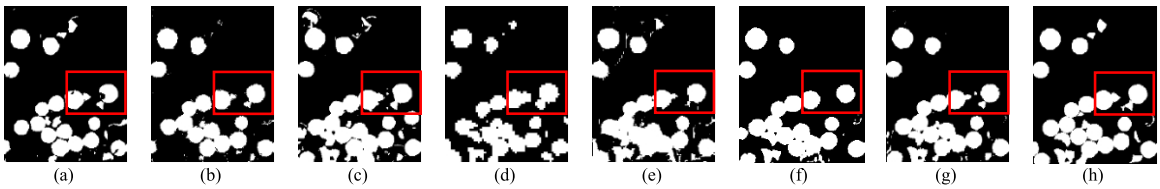


Fig. 9. Binary change maps of different methods on the Benton dataset. (a) ML-EDAN. (b) MSDFFN. (c) SST-Former. (d) D^{2}AGCN. (e) MGCN. (f) GTransCD. (g) CMGA. (h) Ground-truth image.

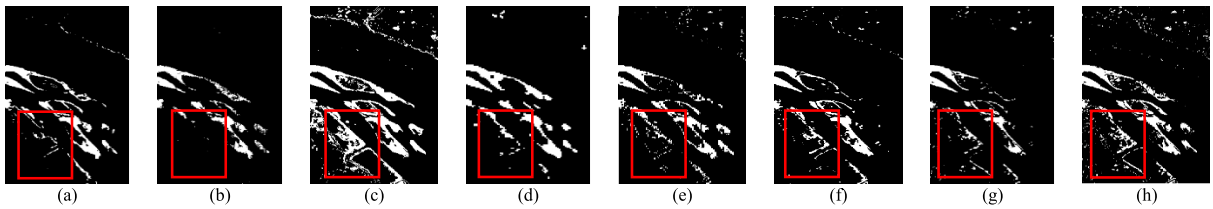


Fig. 10. Binary change maps of different methods on the River dataset. (a) ML-EDAN. (b) MSDFFN. (c) SST-Former. (d) D^{2}AGCN. (e) MGCN. (f) GTransCD. (g) CMGA. (h) Ground-truth image.

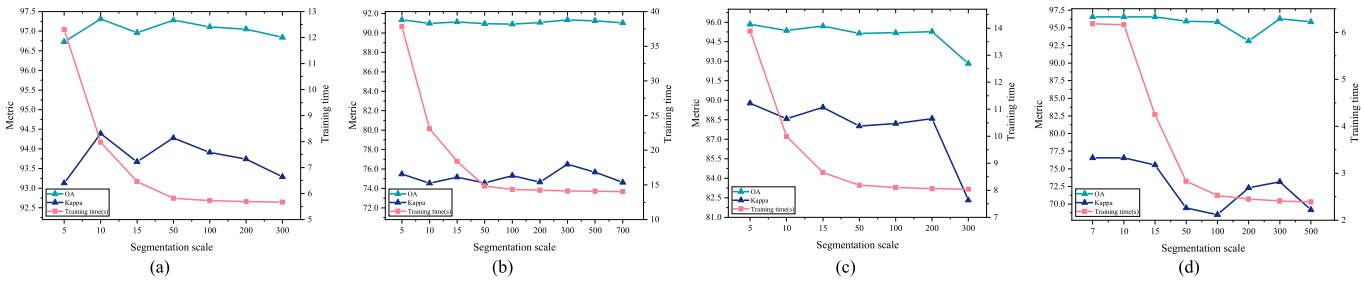


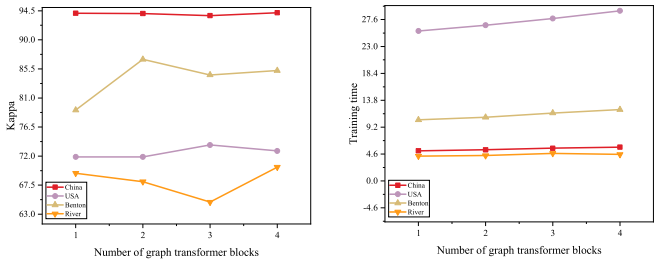
Fig. 11. Performance of CMGA with different segmentation scales on four datasets. (a) China. (b) USA. (c) Benton. (d) River.

we set up the number of GT blocks and multihead attentions are 2.

3) Effect of Training Samples: We utilize different training samples to illustrate the performance of ML-EDAN, MSDFFN, SST-Former, D^{2}AGCN, MGCN, GTransCD, and CMGA. The training samples are set according to the range [0.01%, 0.05%, 0.1%, 0.3%, and 0.5%]. The accuracy of OA and Kappa values of different methods with different training samples is shown in Figs. 14 and 15. The experiments are conducted on the USA dataset for HMCD tasks and the River dataset for HBCD tasks.

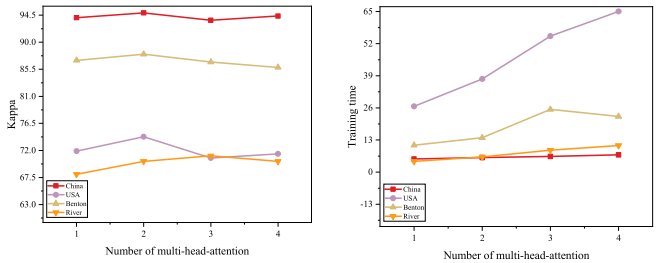
The results show that the performance of all methods increases as the number of training samples increases from 0.01% to 0.5%. CMGA is superior to the comparison methods. Besides, CMGA still obtains the OA value higher than 80 and 90 with the training ratio of 0.01% on the USA dataset and River dataset, respectively. The results confirm the generalization ability of CMGA with limited training samples.

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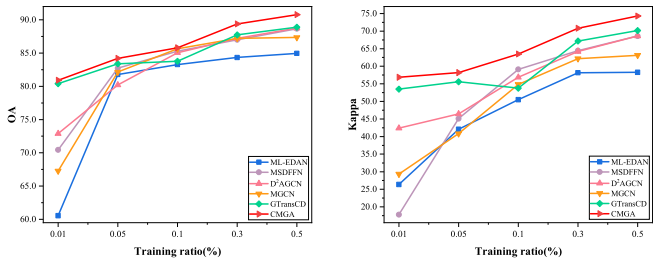
(a) (b)

Fig. 12. Performance of CMGA with different values of L on four datasets. (a) Kappa. (b) Training time.



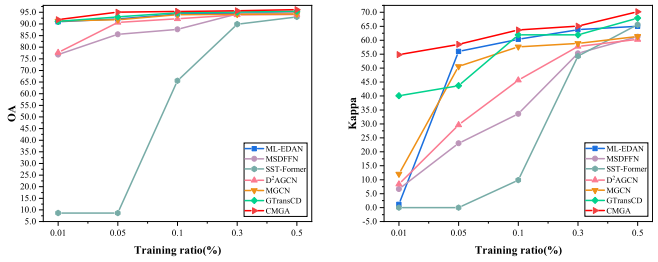
(a) (b)

Fig. 13. Performance of CMGA with different values of c on four datasets. (a) Kappa. (b) Training time.



(a) (b)

Fig. 14. CD results with different training samples on the USA dataset. (a) OA. (b) Kappa.



(a) (b)

Fig. 15. CD results with different training samples on the River dataset. (a) OA. (b) Kappa.

V. Conclusion

In this article, we proposed CMGA to transfer meta-knowledge to guide HBCD and HMCD tasks. The GA module is developed to comprehensively extract features on diverse HSIs with limited samples. To our knowledge, this is the first attempt to leverage optimization-based meta-learning to accomplish both HBCD and HMCD tasks. The model parameters are optimized by various tasks with the same goal. We unify the HBCD task and HMCD task into one framework, and thereby, the optimized model can be effectively transferred to various HCD tasks. Extensive experiments were conducted on four datasets, confirming the effectiveness of CMGA.

However, in order to obtain the general meta-knowledge, we need to pretrain a model on the Indian Pines dataset and the Santa Barbara dataset. Compared with foundation models on large datasets, the model pretrained on small datasets still has limitations in feature representation. Also, a very small number of training samples are still required to adapt the pretrained model to various CD tasks. In the future, we will extend the model to accomplish one-shot and zero-shot learning tasks using foundation models pretrained on extensive datasets.

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